Temperature affects AI responses by controlling the randomness of the output. It does this by adjusting the probability distribution of the next word, or token. Low temperature makes the model more deterministic and conservative, causing it to always choose the most likely token. This makes responses highly predictable, factual, and consistent. On the other hand, a high temperature gives less likely tokens a better chance of being selected. It introduces randomness and diversity.

Top_p, or nucleus sampling, controls more of the vocabulary size which is where the model selects the next token. By selecting a Low Top_p, the model is restricted to a very small set of the most probable tokens and it makes responses focused, coherent and predictable. Whereas, a High Top_p allows the model a much broader range of tokens and includes the less likely ones. It makes the responses more diverse and conversational.